Quiz 1 solutions

Q1

2 Points

Suppose that we fit a linear model to predict Y from variables $X_1, \ldots, X_p$. In order to interpret how X_1 influences the model's *predictions*, the linear model must be a good model of the data.

- ○ True
- ◉ False

We can still look at the coefficients and say how we're making the predictions, even if they're not good predictions!

Q2

2 Points

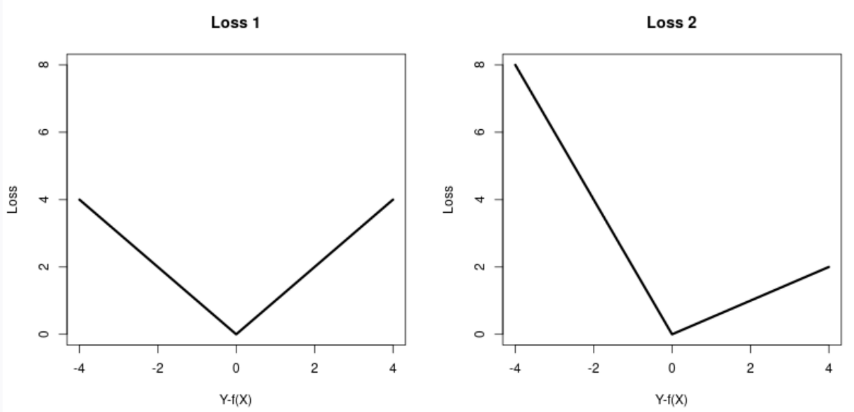
Suppose that we fit a linear model for predicting Y from covariates $X_1, \ldots, X_p$, with $n = 500$ observations and $p = 50$ variables. If we double our sample size, we would expect

- ○ The variance of our predictor to increase
- ◉ The variance of our predictor to decrease
- ○ The bias of our predictor to increase
- ○ The bias of our predictor to decrease

Increasing the sample size decreases the variance, since the variance of linear regression behaves like $\frac{1}{n}$. We would not expect a meaningful change in bias, since that is just a property of the "average" line averaged over all possible data sets.

Q3

2 Points



Consider the two loss functions shown above, as functions of $Y - f(X)$ where $f(X)$ is our prediction function. Let $f_1(X)$ and $f_2(X)$ be the optimal prediction functions for Loss 1 and Loss 2, respectively (assuming that you know the true (X, Y) distribution $\mathcal{P}$). Then

- ○ $f_1(x) < f_2(x)$
- ◉ $f_1(x) > f_2(x)$

Consider Loss 2. It penalizes more for small values of $Y - f(X)$ than large values, in comparison to Loss 1. This means that a predictor that is built to optimize Loss 2 instead of Loss 1 will skew toward larger values of $Y - f(X)$, which means that the corresponding $f(X)$ will have to be smaller than the one for Loss 1.

Q4

3 Points

Suppose that you observe covariates X_1, X_2, X_3 and outcome Y. You fit a linear model $\hat{f}_1$ to predict Y from (X_1, X_2, X_3). You fit another linear model $\hat{f}_2$ to predict Y from $(2X_1, X_2, X_3)$ (note, this scaling will be made on both the training and test data for $\hat{f}_2$). How do the predictions of Y made by $\hat{f}_1$ and $\hat{f}_2$ differ, or are they the same?

They are the same. This is easiest to see by thinking of the fitted values from regression as a projection. Rescaling one of the columns of the X matrix by 2 does not change the span of the columns of X, so does not change the projection onto it.

Q5

3 Points

Suppose that the true data generating process is linear, $Y = X\beta + \epsilon$ (where $X \in \mathbb{R}^{n\times p}$ and $\beta, \epsilon \in \mathbb{R}^p$), with a non-zero coefficient vector and a greater number of observations than features. You consider two different approaches to prediction:
a. Fit a least-squares linear regression on the training data, and apply it to the test data.
b. Simply take the mean of Y in the training data, and guess this constant value for all test points.

Explain a scenario where approach b will have lower test set MSE than approach a, or explain why such a scenario does not exist.

If the strength of the predictors is small and the dimension is large, approach b can do better in test error. That's because the bias introduced by just setting all the coefficients to zero will be small, and the variance reduction from not estimating the coefficients will be large.

Q6

3 Points

Suppose that you are predicting housing prices Y from covariates X, and that you care about the *mean squared relative error* in your predictions:

$$MSRE = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{Y_i - f(X_i)}{Y_i}\right)^2$$

However, you don't know how to fit a model to minimize that loss, so you fit simple least squares regression. How do you expect your least squares predictions to differ from a linear model that minimized the MSRE directly?

The MSRE has a lower penalty on observations with higher price Y_i. This means that a model that optimizes MSRE will tend to do worse on the higher price houses than a model that optimizes MSE (and therefore focuses more on the large Y_i).

You can also see this by thinking of the $1/Y_i^2$ as regression weights, if you happen to be familiar.